

Shamit Fatin

<https://shamit187.github.io>
GitHub: Shamit187

shamit.f@utah.com
+1 (385) 774-2507 (US)
+8801842227053 (BD)

PROFILE

I am a PhD student at the University of Utah (Kahlert School of Computing), advised by Professor Anna Fariha. My research focuses on data management, particularly semantic data change summarization.

EDUCATION

- **University of Utah (U of U)** Salt Lake City, Utah, USA
PhD in Computer Science (Data Management) Aug 2025 – Present
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
BSc in Computer Science and Engineering (Honours), CGPA: 3.90 Apr 2019 – Jul 2024

RESEARCH EXPERIENCE

- **Semantic Data Change Summarization** Salt Lake City, Utah
PhD Research Project 2025 – Present
Developing interpretable, scalable methods for summarizing accurately and interpretably how structured datasets evolve over time
 - **Advisor:** Anna Fariha
- **SystemVerilog Code Generation using LLMs** Dhaka, Bangladesh / Remote
ML Engineer 2025
Building LLM driven agentic pipelines for generating synthesizable SystemVerilog, automating testbench creation, module templates, and sequential logic behavior using fine tuned models.
 - **Advisor:** Anindya Iqbal
 - **Collaborator:** Aszad Rakin
 - **Sponsor:** Asian Development Bank
- **LELANTE: LLM-based Automated Android Testing** Dhaka, Bangladesh
Lead Researcher 2024 – 2025
Designed a multimodal LLM based framework for automated Android UI testing
 - **Advisor:** Anindya Iqbal, Sukarna Barua
 - **Collaborator:** HAZ Sameen Shahgir, Mehbubul Hasan Al-Quvi
 - **Sponsor:** Samsung Research
 - **Manuscript:** EASE 2024
- **COMPASS: Geo-Spatial Reasoning in LLMs** Dhaka, Bangladesh
Undergraduate Thesis 2023 – 2024
Explored the use of LLMs for geo-spatial reasoning tasks such as popular-path prediction, showing significant improvements over traditional heuristics using foundational models.
 - **Advisor:** Mohammed Eunus Ali, Md Rizwan Parvez
 - **Collaborator:** Nazmul Islam Ananto
 - **Manuscript:** EACL 2025

RESEARCH PAPERS

- **LELANTE: LEveraging LLM for Automated ANDroid TESting**
EASE 2025: Emerging Results
We introduce LELANTE, a framework that automates Android test execution directly from natural language descriptions without requiring Appium/Espresso scripts. LELANTE iteratively interprets instructions, generates action plans using chain-of-thought reasoning, and executes actions on-device through a GUI controller. The system integrates screen refinement, structured prompting, and LLM distillation to reduce cost and improve scalability. Across 390 test cases from 10 popular apps, LELANTE achieves a 73% success rate, showing that LLMs can effectively bridge natural-language test specifications and automated mobile testing.

Co-authors: Shamit Fatin, HAZ Sameen Shahgir, Sukarna Barua, Anindya Iqbal

- **CompassLLM: A Multi-Agent Approach toward Geo-Spatial Reasoning for Popular Path Query**
EACL (Under Review)

COMPASS introduces a multi-agent LLM-based framework for geo-spatial reasoning, particularly the popular-path problem. The system combines a “Search” phase that identifies candidate popular routes with a “Generate” phase that synthesizes novel paths under user-defined constraints. By leveraging LLMs’ ability to interpret spatial relationships and graph structure, COMPASS overcomes the limitations of classical algorithms, especially in sparse or constrained scenarios. Experiments on real and synthetic trajectory datasets show strong performance in both path retrieval and novel path generation.

Co-authors: Nazmul Islam Ananto, Shamit Fatin, Md Rizwan Parvez, Mohammed Eunus Ali